Fizika

9-sinf

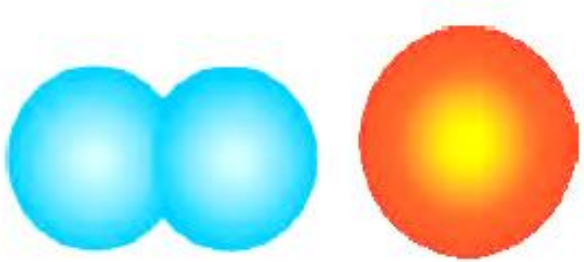
Mavzu: **Yoqilg‘ining solishtirma yonish issiqligi**



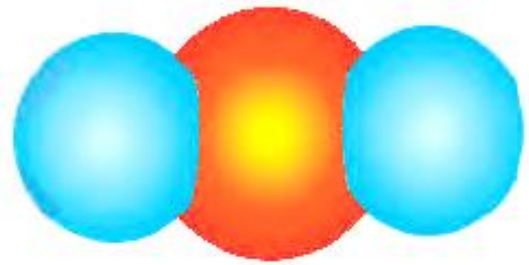





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1 kg yonilg‘i batamom yonganda undan ajralib chiqadigan issiqlik miqdoriga solishtirma yonish issiqligi deyiladi.

$$Q = qm \qquad q = \frac{Q}{m}$$

$$[q] = \left[\frac{Q}{m}\right] = \frac{1\ J}{1\ kg} = 1\ \frac{J}{kg}$$

	Yoqilg‘i	Solishtirma yonish issiqligi, (MJ/kg)		Yoqilg‘i	Solishtirma yonish issiqligi, (MJ/kg)
1	Benzin	46	4	Quruq o‘tin	10
2	Kerosin	42	5	Tabiiy gaz	44
3	Toshko‘mir	29	6	Spirt	29

1-namunaviy masala 66-bet

Massasi 20 kg bo‘lgan toshko‘mir yonganda ajralib chiqaradigan issiqlikni olish uchun, qancha quruq o‘tinni yoqish kerak?

Berilgan:

$m_1 = 20kg$

$q_1 = 29\frac{MJ}{kg} = 29 \cdot 10^6 J/kg$

$q_2 = 10\frac{MJ}{kg} = 10 \cdot 10^6 J/kg$

$Q_1 = Q_2$

Formula:

$Q = qm$ dan

$Q_1 = q_1 m_1 \quad Q_2 = q_2 m_2$

$Q_1 = Q_2$ dan

$q_1 m_1 = q_2 m_2 \Rightarrow m_2 = \frac{m_1 q_1}{q_2}$

Topish kerak: $m_2 - ?$

Yechish: $m_2 = \frac{20\ kg \cdot 29 \cdot 10^6\ J/kg}{10 \cdot 10^6\ J/kg} = 58\ kg$

Javob: 58 kg quruq o‘tin yoqish kerak.

16-mashq 1-masala

Massasi qanday bo‘lgan spirt yonganda 5,8 MJ issiqlik ajralib chiqadi?

Spirtning solishtirma yonish issiqligi $\mathbf{2,9 \cdot 10^7 J/kg}$.

Berilgan:

$$Q = 5{,}8\,\frac{MJ}{kg} = 5{,}8 \cdot 10^6\,\frac{J}{kg}$$

$$q = 2{,}9 \cdot 10^7\,\frac{J}{kg}$$

Topish kerak: $m - ?$

Formula:

$$Q = qm$$

$$m = \frac{Q}{q}$$

Yechish:

$$m = \frac{5{,}8 \cdot 10^6\,\frac{J}{kg}}{2{,}9 \cdot 10^7\,\frac{J}{kg}} = 0{,}2\ kg = 200\ g$$

Javob: 200 g spirt yonganda.

16-mashq 3-masala

Neksiya avtomobili har 100 km da o‘rtacha 10 l benzin sarflasa, har bir kilometrda qancha issiqlik ajralib chiqadi?

Benzinning zichligi $\boldsymbol{700\,\frac{kg}{m^3}}$.

Berilgan:

$S = 100\ km$

$V = 10\ l = 0{,}01\ m^3$

$\Delta S = 1\ km$

$\rho = 700\ \frac{kg}{m^3}$

$q = 46\ \frac{Mj}{kg} = 46 \cdot 10^6\ \frac{J}{kg}$

Topish kerak: $\Delta Q - ?$

Formula:

$$m = \rho V$$

$$Q = qm$$

$$\Delta Q = \frac{\Delta S}{S} \cdot Q$$

Yechish:

$$m = 700\,\frac{kg}{m^3} \cdot 0{,}01\,m^3 = 7\,kg$$

$$Q = 7\,kg \cdot 46 \cdot 10^6\,\frac{J}{kg} = 322 \cdot 10^6 J$$

$$\Delta Q = \frac{1\,km}{100\,km} \cdot 322 \cdot 10^4 J = 3{,}22 \cdot 10^6\,J$$

Javob: $3,22 \cdot 10^6\,J$ issiqlik ajralib chiqadi.

Mustaqil bajarish uchun topshiriqlar.

1. Mavzuni o‘qib, mavzuga doir savollarga javob yozish. (66-bet)

2. 16-mashq 2-4-masalalarni yechish. (66-bet)